

CREDIT DATASET (ISLR) · 400 CUSTOMERS

What income, age and balance tell us us about our customers

Business implications of three descriptive analyses

SCOPE

What these statistics can and cannot tell us

All three analyses are **univariate**: each one describes a single variable on its own.

They tell us

What our customer base looks like in terms of income, age and balance, each taken separately.

They do not tell us

How the variables are related, for example whether higher earners carry larger balances.

The implications that follow are observations and hypotheses to guide decisions, not proven cause-and-effect relationships.

INCOME

Income: a long tail of high earners

\$40,672

Mean

the arithmetic average

\$29,715

Median

half earn less, half earn more

\$35,524

Standard deviation

almost as large as the mean

1.76

Skewness

positive: a long right tail

The mean sits about **\$11,000 above the median**: a small group of high earners pulls the average up.

INCOME · IMPLICATION 1

The “average customer” does not really exist

\$29,715

Median: the typical customer

\$40,672

Mean: inflated by a few high earners

Planning around the mean would overshoot the financial capacity of most customers. For targets, pricing and product design aimed at the mass market, we use the **median** as the reference point.

A heterogeneous base calls for segmentation

Mass-market segment

The large majority, with low-to-medium incomes

Simple, low-cost, standardised products and efficient digital channels.

Affluent segment

A small group of high earners in the long right tail

Dedicated relationship managers, premium products and wealth-management services.

Hypothesis to test

A small share of customers may generate a large share of balances or revenue. If account-level data confirm it, retaining the high-income minority becomes a priority.

Credit policy and data quality

Decide on individual income, not averages

A limit that is prudent at \$30,000 is needlessly restrictive at \$150,000, and vice versa.

Affordability checks and credit limits should use each customer's own income; risk models should account for the skewed shape, for example with a log transformation.

\$12

A minimum income that is almost certainly an error

A wrong income can decline a creditworthy customer or pass an affordability check on false grounds. Validation rules at data entry protect customers and the bank.

AGE

Age: symmetric and evenly spread

55.7

Mean age

in years

56

Median

practically equal to the
mean

23–98

Range

75 years between youngest
and oldest

–1.07

Kurtosis

flat shape, light tails

Customers are spread **fairly evenly across all adult ages**, with no dominant generation.

Our customers span the whole life cycle

Young adults

- First current accounts
- Credit cards and personal loans
- Digital-first services

Middle age

- Mortgages
- Family protection
- Savings and investment plans

Older customers

- Retirement income
- Wealth preservation
- Estate and succession planning

Every life stage is represented in meaningful numbers: a strategy focused on one age group would leave large parts of the base underserved.

A relatively mature customer base

Half of our customers are over 56.

Wealth transfer

Assets may pass to heirs in the coming years: a risk of outflows and a chance to engage the next generation.

Balanced channels

Older customers may prefer branches or phone support; younger ones expect digital. Neither can be neglected.

Vulnerable customers

Customers in their late 80s and 90s may need clear communication, accessible services and stronger fraud protection.

A gap at the young end, and a clean variable

No customers under 23

It may reflect eligibility rules, how the data were collected, or a genuine acquisition gap.

Customers acquired young tend to stay for years: student and first-job offers could help renew an ageing base.

Age is clean and reliable

No outliers: both extremes lie within about 2.5 standard deviations of the mean.

Symmetric shape: usable in reports and models without transformation, and a dependable basis for segmentation.

BALANCE

Balance: many customers owe nothing

\$520

Mean

average card balance

\$460

Median

slightly below the mean

\$0

Mode

the most common value

0.58

Skewness

a mild right tail

Two groups: **a cluster of customers at zero** and a wide spread of balances up to \$1,999.

Two groups call for different strategies

Zero-balance customers

Pay in full each month or rarely use the card

Little interest revenue: usage incentives and fee-based services fit them.

Customers carrying a balance

The source of interest income and credit risk

Credit monitoring, instalment plans and responsible-lending checks.

Hypothesis to test

Balance may depend on income and credit limit. If high balances concentrate among low-income customers, affordability risk rises and needs closer monitoring.

Credit risk and data quality

Watch the high end: credit risk

The maximum of \$1,999 lies about 3.2 standard deviations above the mean: a borderline outlier.

A balance close to the credit limit can signal financial stress: comparing Balance with Limit shows how close each customer is to the maximum allowed.

\$0

Are the zeros real, or missing values?

Missing values recorded as 0 would distort every statistic. Verifying the zeros protects reports, models and credit decisions.

Reporting and modelling the zeros

Report the two groups separately

One overall average mixes customers who owe nothing with those who do.

We count the zeros and describe customers with a positive balance on their own, for example with AVERAGEIF.

Models must handle the zeros

A plain linear regression may predict negative balances, which are impossible.

A two-part approach first predicts whether a customer carries a balance, then how much.

PRECISION

How much can we trust the averages?

±3%

Age

±1.7 years on a mean of 55.7:
estimated very precisely.

±9%

Income

±\$3,500 on a mean of \$40,700:
much less precise, because of
high variability.

±9%

Balance

±\$45 on a mean of \$520:
moderate precision, limited by
high variability.

Relative margin of error = margin of error ÷ mean. Targets and forecasts based on average income or balance carry more uncertainty, and planning should reflect it.

NEXT STEPS

Study the variables together

01

Relationship

Scatter plots and the correlation coefficient (from -1 to $+1$) show, for example, whether balance rises with income.

02

Age $\times$ income segments

A cross-table of age bands and income bands reveals combined groups, such as young high earners or older low-income customers.

03

Link to outcomes

Adding balances, product holdings and credit behaviour turns who our customers are into which customers create value or risk.